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[Am J Chin Med.](#) 2010;38(6):1093-106. doi: 10.1142/S0192415X10008494.

Protective effects on mitochondria and anti-aging activity of polysaccharides from cultivated fruiting bodies of *Cordyceps militaris*

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PMID: 21061463 DOI: [10.1142/S0192415X10008494](#)

Abstract

Cordyceps militaris (L.) Link is an entomopathogenic fungus parasitic to Lepidoptera larvae, and is widely used as a folk tonic or invigorant for longevity in China. Although *C. militaris* has been used in traditional Chinese medicine for millennia, there is still a lack convincing evidence for its anti-aging activities. This study was performed to investigate the effects of polysaccharides from cultivated fruiting bodies of *C. militaris* (CMP) on mitochondrial injury, antioxidation and anti-aging activity. Fruiting bodies of *C. militaris* were cultivated artificially under optimized conditions. The spectrophotometric method was used to measure thiobarbituric acid reactive substances (TBARS), mitochondrial swelling, and activities of scavenging superoxide anions in vitro. D-galactose (100 mg/kg/day) was injected subcutaneously into back of the neck of mice for 7 weeks to induce an aging model. The effects of CMP on the activities of catalase (CAT), superoxide dismutase (SOD), glutathione peroxidase (GPx) and anti-hydroxyl radicals were assayed in vivo using commercial monitoring kits. The results showed that CMP could inhibit mitochondrial injury and swelling induced by Fe²⁺(+)-L-Cysteine in a concentration- dependent manner and it also had a significant superoxide anion scavenging effect. Moreover, the activities of CAT, SOD, GPx and anti-hydroxyl radicals in mice liver were increased significantly by CMP. These results indicate that CMP protects mitochondria by scavenging reactive oxygen species (ROS), inhibiting mitochondrial swelling, and increasing the activities of antioxidant enzymes. Therefore, CMP may have pharmaceutical values for mitochondrial protection and anti-aging. CMP was the major bioactive component in *C. militaris*.

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